

IE-02: Ideal and worst-case GMRES coincide for a Jordan block

George Stepaniants

Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, USA.

This manuscript proves the full complex Jordan-block equality in IE-02. It was developed with substantial ChatGPT/Codex assistance at the author's request. A separate [independent Codex-agent review](#) returned PASS for the complete target. This is automated-agent review; no external human peer review or formal proof-assistant verification is asserted.

The argument proves a stronger statement for finite affine families of triangular Toeplitz matrices. It uses finite Carathéodory-Fejér interpolation and scalar polynomial spectral factorization, with no simplicity, divisibility, or eigenvalue-regime restriction.

1. Exact target and notation

Theorem 1 (IE-02). Let $n \geq 2$, let $\lambda \in \mathbb{C} \setminus \{0\}$, and let $1 \leq k < n$. If $J_n(\lambda) = \lambda I + N$, where $N_{i,i+1} = 1$ and all other entries are zero, and

$$\mathcal{P}_k = \{p \in \mathbb{C}[z] : \deg p \leq k, p(0) = 1\},$$

then

$$\max_{\|v\|_2=1} \min_{p \in \mathcal{P}_k} \|p(J_n(\lambda))v\|_2 = \min_{p \in \mathcal{P}_k} \|p(J_n(\lambda))\|_2. \quad (1)$$

All vectors and polynomials are complex. We identify $\mathbb{C}^n$ isometrically with

$$\mathcal{H}_n = \{f \in \mathbb{C}[z] : \deg f \leq n-1\} \subset L^2(\mathbb{T}, d\theta/(2\pi)),$$

using the coefficient basis $1, z, \dots, z^{n-1}$. Inner products are $\langle f, g \rangle = \int_{\mathbb{T}} \bar{f}g d\theta/(2\pi)$, linear in the second argument. Write Π_n for the orthogonal projection onto $\mathcal{H}_n$. Every lower triangular Toeplitz matrix has a unique symbol $r \in \mathcal{H}_n$ and acts as

$$Rf = \Pi_n(rf), \quad f \in \mathcal{H}_n. \quad (2)$$

2. The maximal singular subspace of a triangular Toeplitz matrix

We use the following classical finite Carathéodory-Fejér theorem, in exactly the form stated as **Theorem CF, §2, p. 84** of Courtney and Sarason [CS12]: for every nonzero lower triangular Toeplitz matrix T of size n , if $t = \|T\|_2$, there is a finite Blaschke product B of order $d \leq n-1$ whose multiplication operator compresses to T/t on $\mathcal{H}_n$. Thus

$$B(z) = \gamma \prod_{j=1}^d \frac{z - \alpha_j}{1 - \bar{\alpha}_j z} = \frac{a(z)}{b(z)}, \quad |\gamma| = 1, \quad |\alpha_j| < 1, \quad (3)$$

where

$$a(z) = \gamma \prod_{j=1}^d (z - \alpha_j), \quad b(z) = \prod_{j=1}^d (1 - \bar{\alpha}_j z).$$

Empty products are allowed. In particular $\deg a = d$, $\deg b \leq d$, a and b are relatively prime, b has no zeros on the closed unit disk, and $|a| = |b|$ on $\mathbb{T}$. The theorem says

$$Tf = t\Pi_n(Bf) \quad (f \in \mathcal{H}_n). \quad (4)$$

Lemma 2. The maximal right singular subspace of T is

$$E := \ker(T^*T - t^2I) = \{bh : h \in \mathbb{C}[z], \deg h \leq n - 1 - d\}. \quad (5)$$

Moreover, for every such h ,

$$T(bh) = tah. \quad (6)$$

Proof. Multiplication by B is an isometry on $L^2(\mathbb{T})$, since $|B| = 1$ there. By (4) and the equality case for an orthogonal projection,

$$\|Tf\|_2 = t\|f\|_2 \iff Bf \in \mathcal{H}_n.$$

This norm equality is equivalent to $f \in E$, because $t^2I - T^*T$ is positive semidefinite. If $Bf = g \in \mathcal{H}_n$, then $af = bg$ as polynomial identities. Relative primeness gives $f = bh$ and $g = ah$ for a polynomial h . Since $\deg a = d$, the condition $g \in \mathcal{H}_n$ gives $\deg h \leq n - 1 - d$ (with zero vectors included). Conversely, that degree bound gives both $bh, ah \in \mathcal{H}_n$ and $B(bh) = ah$, proving (5) and (6). $\square$

3. Replacing a convex combination by a single vector

We recall a scalar factorization fact and include its algebraic proof.

Lemma 3 (polynomial spectral factorization). For finitely many complex polynomials $h_1, \dots, h_s$ of degree at most m , and nonnegative real numbers $\omega_1, \dots, \omega_s$, there is a complex polynomial h of degree at most m such that

$$|h(e^{i\theta})|^2 = \sum_{\nu=1}^s \omega_\nu |h_\nu(e^{i\theta})|^2 \quad \text{for all real } \theta. \quad (7)$$

Proof. The right side is a nonnegative trigonometric polynomial Q of degree at most m . If it is zero or constant the assertion is immediate. Otherwise let its effective degree be $\ell \leq m$. The ordinary polynomial $z^\ell Q(z)$ has degree 2ℓ , nonzero constant term, and roots paired under $\zeta \mapsto 1/\bar{\zeta}$. Every root on the unit circle has even multiplicity, since the real analytic function $Q(e^{i\theta})$ is nonnegative. Choose one root from each pair off the circle, with multiplicity, and half of each multiplicity on the circle. The polynomial h_0 with these ℓ roots satisfies $Q = \beta |h_0|^2$ on the circle for a constant β . Both sides are positive at some point, so $\beta > 0$. Set $h = \sqrt{\beta} h_0$. $\square$

Lemma 4 (simultaneous preservation). Let $T \neq 0$ be lower triangular Toeplitz with norm t , and let $R_1, \dots, R_k$ be lower triangular Toeplitz matrices of the same size. Suppose $f_1, \dots, f_s$ are unit vectors in the maximal right singular subspace E of T , and $\omega_\nu \geq 0$, $\sum_\nu \omega_\nu = 1$. There is a unit vector $f \in E$ such that, simultaneously for $j = 1, \dots, k$,

$$\langle Tf, R_j f \rangle = \sum_{\nu=1}^s \omega_\nu \langle Tf_\nu, R_j f_\nu \rangle. \quad (8)$$

Proof. Write $f_\nu = b h_\nu$ as in Lemma 2, so all h_ν have degree at most $m = n - 1 - d$. Apply Lemma 3 to obtain h satisfying (7), and put $f = b h$. Lemma 2 shows $f \in E$ and $Tf = t a h$. Furthermore,

$$\|f\|_2^2 = \int_{\mathbb{T}} |b|^2 |h|^2 = \sum_\nu \omega_\nu \int_{\mathbb{T}} |b|^2 |h_\nu|^2 = 1. \quad (9)$$

Here and below the normalized circle measure is understood. If r_j is the symbol in (2) for R_j , then $a h \in \mathcal{H}_n$ allows the orthogonal projection to be removed from the second argument, giving

$$\begin{aligned} \langle Tf, R_j f \rangle &= t \langle a h, \Pi_n(r_j b h) \rangle \\ &= t \int_{\mathbb{T}} \bar{a} r_j b |h|^2 \\ &= \sum_\nu \omega_\nu t \int_{\mathbb{T}} \bar{a} r_j b |h_\nu|^2 \\ &= \sum_\nu \omega_\nu \langle T f_\nu, R_j f_\nu \rangle. \end{aligned} \quad (10)$$

This proves (8) as a complex equality, preserving both real and imaginary parts. $\square$

4. The finite-dimensional optimality condition

Lemma 5. Suppose a nonzero matrix T minimizes the operator norm on the complex affine space $T + \text{span}_{\mathbb{C}}\{R_1, \dots, R_k\}$. Put $t = \|T\|_2$ and $E = \ker(T^*T - t^2 I)$. Then finitely many unit vectors $f_\nu \in E$ and weights $\omega_\nu \geq 0$ with $\sum_\nu \omega_\nu = 1$ satisfy

$$\sum_\nu \omega_\nu \langle T f_\nu, R_j f_\nu \rangle = 0 \quad (j = 1, \dots, k). \quad (11)$$

Proof. Consider the compact set

$$\mathcal{G} = \{(\langle Tf, R_1 f \rangle, \dots, \langle Tf, R_k f \rangle) : f \in E, \|f\|_2 = 1\} \subset \mathbb{C}^k \cong \mathbb{R}^{2k}.$$

If zero is outside its convex hull, finite-dimensional strict separation provides complex scalars c_j and $\gamma > 0$ such that, with $D = \sum_j c_j R_j$,

$$\text{Re}\langle Tf, Df \rangle \geq \gamma \quad (f \in E, \|f\|_2 = 1). \quad (12)$$

Continuity gives a neighborhood U of the unit sphere of E in the full unit sphere on which the left side is at least $\gamma/2$. On the compact complement of U , there is $\beta > 0$ such that $\|Tf\|_2^2 \leq t^2 - \beta$; if the complement is empty, that part of the argument is unnecessary. For a sufficiently small real $\varepsilon > 0$,

$$\|(T - \varepsilon D)f\|_2^2 = \|Tf\|_2^2 - 2\varepsilon \operatorname{Re}\langle Tf, Df \rangle + \varepsilon^2 \|Df\|_2^2 < t^2$$

uniformly on U , by the $\gamma/2$ bound, and uniformly on its complement, by the β bound and boundedness of all other terms. This contradicts minimality of T . Thus zero is in the convex hull of $\mathcal{G}$. The definition of the convex hull (or Carathéodory's theorem) now gives a finite convex combination yielding (11). $\square$

5. Completion of the proof

Theorem 6 (triangular Toeplitz affine minimax). Let $Y, R_1, \dots, R_k$ be complex lower triangular Toeplitz matrices of the same size, and let $\mathcal{X} = \operatorname{span}_{\mathbb{C}}\{R_1, \dots, R_k\}$. Then

$$\max_{\|f\|_2=1} \min_{X \in \mathcal{X}} \|(Y - X)f\|_2 = \min_{X \in \mathcal{X}} \|Y - X\|_2. \quad (13)$$

Proof. The affine matrix space $Y - \mathcal{X}$ is closed and finite dimensional, so its distance to zero is attained, say at $T = Y - X_*$. If $T = 0$, both sides of (13) are zero. Otherwise apply Lemma 5 to this minimizer, and then Lemma 4 to its convex combination. The resulting unit vector f satisfies

$$\|Tf\|_2 = \|T\|_2 = t, \quad \langle Tf, R_j f \rangle = 0 \quad (j = 1, \dots, k). \quad (14)$$

For every $X \in \mathcal{X}$, the difference $X_* - X$ lies in $\mathcal{X}$; hence orthogonality and the Pythagorean identity give

$$\|(Y - X)f\|_2^2 = \|Tf + (X_* - X)f\|_2^2 = t^2 + \|(X_* - X)f\|_2^2 \geq t^2. \quad (15)$$

Taking $X = X_*$ attains equality. Thus the left side of (13) is at least t , while it is at most the right side by the operator norm bound. $\square$

Proof of Theorem 1. Let W be the unitary reversal matrix. Then $A = WJ_n(\lambda)W^* = \lambda I + S$, where S is the lower shift. All $I, A, A^2, \dots, A^k$ are lower triangular Toeplitz. Apply Theorem 6 with $Y = I$ and $\mathcal{X} = \operatorname{span}_{\mathbb{C}}\{A, \dots, A^k\}$. The matrices $Y - X$ are exactly $p(A)$ for $p \in \mathcal{P}_k$. Unitary invariance and the bijection of unit vectors induced by W give (1). This applies directly to every allowed complex λ , without a phase reduction. $\square$

For the stated target the minimum is strictly positive: if $p(A) = 0$ and $\deg p < n$, the independent powers $I, S, \dots, S^{n-1}$ show that all Taylor coefficients of $p(\lambda + z)$ vanish, forcing $p = 0$ and contradicting $p(0) = 1$. The proof nevertheless already treated a zero minimum in the more general affine theorem. It also treats constant Blaschke products, singular residual matrices, maximal singular values of any multiplicity, and stagnation without separate assumptions.

Scope and verification

Theorem 1 establishes the exact original target for all dimensions, every allowed degree, all nonzero complex eigenvalues, and complex polynomials and starting vectors. Theorem 6 also proves the affine minimax equality for triangular Toeplitz matrices. The proof does not change the matrix or use a doubled dimension.

The classical finite Carathéodory-Fejér theorem is the only external analytic input. The scalar factorization, maximal singular-subspace description, finite-dimensional optimality condition, and preservation of all complex orthogonality equations are proved here. The [submission record](#) preserves the exact research source and documents independent review and source-conversion checks.

References

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